C.I.: Illustrative Examples

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Example 7.2-1

In the preceding discussion, let $n = 15$, $m = 8$, $\bar{x} = 70.1$, $\bar{y} = 75.3$, $\sigma_X^2 = 60$, $\sigma_Y^2 = 40$, and $1 - \alpha = 0.90$. Thus, $1 - \alpha/2 = 0.95 = \Phi(1.645)$. Hence,

$$1.645\sigma_W = 1.645\sqrt{\frac{60}{15} + \frac{40}{8}} = 4.935,$$

and, since $\bar{x} - \bar{y} = -5.2$, it follows that

$$[-5.2 - 4.935, -5.2 + 4.935] = [-10.135, -0.265]$$

is a 90% confidence interval for $\mu_X - \mu_Y$. Because the confidence interval does not include zero, we suspect that μ_Y is greater than μ_X. ■

Source: Section 7.2 of Textbook

Example 7.2-2

Suppose that scores on a standardized test in mathematics taken by students from large and small high schools are $N(\mu_X, \sigma^2)$ and $N(\mu_Y, \sigma^2)$, respectively, where σ^2 is unknown. If a random sample of $n = 9$ students from large high schools yielded $\bar{x} = 81.31$, $s_x^2 = 60.76$, and a random sample of $m = 15$ students from small high schools yielded $\bar{y} = 78.61$, $s_y^2 = 48.24$, then the endpoints for a 95% confidence interval for $\mu_X - \mu_Y$ are given by

$$81.31 - 78.61 \pm 2.074\sqrt{\frac{8(60.76) + 14(48.24)}{22}}\sqrt{\frac{1}{9} + \frac{1}{15}}$$

because $t_{0.025}(22) = 2.074$. The 95% confidence interval is $[-3.65, 9.05]$. ■

Source: Section 7.2 of Textbook

Example 7.2-4

An experiment was conducted to compare people's reaction times to a red light versus a green light. When signaled with either the red or the green light, the subject was asked to hit a switch to turn off the light. When the switch was hit, a clock was turned off and the reaction time in seconds was recorded. The following results give the reaction times for eight subjects:

Source: Section 7.2 of Textbook

Subject	Red (x)	Green (y)	$d = x - y$
1	0.30	0.43	–0.13
2	0.23	0.32	–0.09
3	0.41	0.58	–0.17
4	0.53	0.46	0.07
5	0.24	0.27	–0.03
6	0.36	0.41	–0.05
7	0.38	0.38	0.00
8	0.51	0.61	–0.10

Source: Example 7.2-4 of Textbook

Practice: Reading statistical table

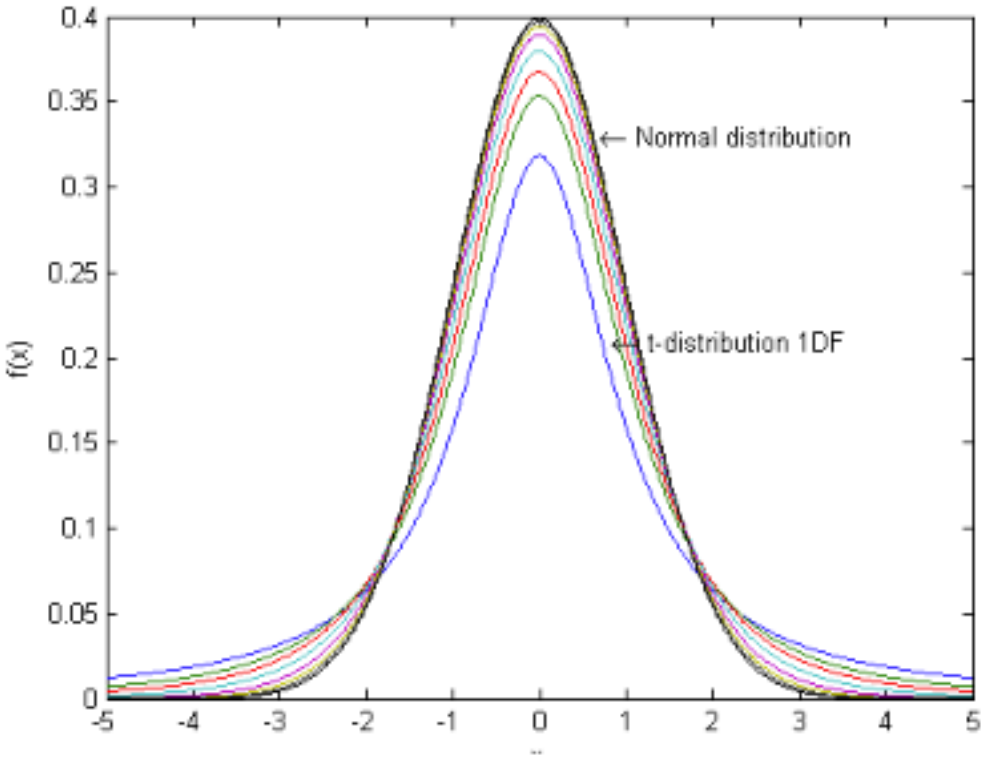
For these data, $\bar{d} = -0.0625$ and $s_d = 0.0765$. To form a 95% confidence interval for $\mu_D = \mu_X - \mu_Y$, we find, from Table VI in Appendix B, that $t_{0.025}(7) = 2.365$. Thus, the endpoints for the confidence interval are

$$-0.0625 \pm 2.365 \frac{0.0765}{\sqrt{8}}, \qquad \text{or} \qquad [-0.1265, 0.0015].$$

In this very limited data set, zero is included in the confidence interval but is close to the endpoint 0.0015. We suspect that if more data were taken, zero might not be included in the confidence interval. If that actually were to happen, it would seem that people react faster to a red light. ■

Source: Example 7.2-4 of Textbook

Student's t-distribution – plots of 1,2,3,4,5,10,20,50 DF and Normal (limiting distribution)



Question: How to show that the **limiting distribution** of t-distribution is standard normal distribution?

Hint: Using your knowledge in a **calculus** course can achieve the desired results.

Handwritten notes can be prepared after class, if students have interests.